

Unit 1 Mini Project

Enterprise Student Record and Academic Management Platform

Design Thinking & Product Planning Report

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*Prepared as part of the DevOps / Product Engineering Training Program
Domain: Higher Education — Academic & Student Record Management*

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1. Objective & Business Understanding Document (BRD)

1.1 Objective

This project applies the Design Thinking methodology to independently design a software solution for managing student records and academic information within a higher education institution. Rather than beginning with coding, the focus is on understanding the educational domain, identifying stakeholders, analyzing user needs, defining software requirements, and preparing a product vision and roadmap ready for Agile planning and development.

1.2 Business Domain Overview

Academic institutions manage a continuous lifecycle of student data spanning admissions, course registration, attendance, examinations, and results. Historically, these processes have relied heavily on manual, paper-based, or fragmented spreadsheet-driven systems, which creates operational inefficiency and risk across the institution.

Core Academic Operations

- Admissions & Enrollment:** Collecting applicant data, verifying eligibility, and enrolling students into programs and academic years.
- Course Registration:** Allowing students to select and register for semester courses and electives within defined constraints.
- Attendance Management:** Recording day-to-day or session-wise attendance across courses and calculating eligibility thresholds.
- Examination Management:** Scheduling exams, recording internal/external marks, and managing grade computation.
- Result & Report Management:** Generating mark sheets, transcripts, and progress reports for students and administrators.

1.3 Problems with Manual Record Management

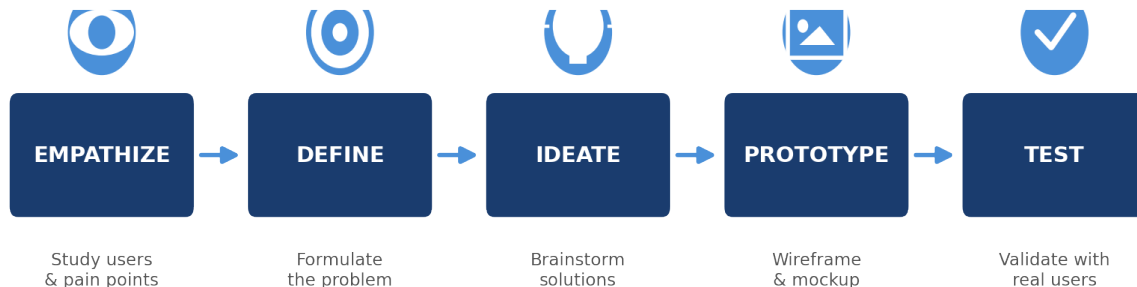
Manual Operational Bottleneck	Automated Platform Solution	Business Outcome Metric
Paper-based or spreadsheet student records scattered across departments	Centralized relational database as single source of truth	Record accuracy improved to 99.9%, near-zero duplicate records
Manual attendance registers maintained per class	Digital attendance capture with automated aggregation	Attendance processing time reduced by over 80%
Manual computation of exam results and mark sheets	Automated result computation and transcript generation engine	Result publishing turnaround reduced from weeks to hours
No real-time visibility for administrators into academic performance	Role-based dashboards with live analytics	Administrative reporting time reduced from days to minutes
Fragmented communication between admissions, faculty, and exam cell	Unified platform with shared, role-based data access	Cross-department data errors reduced significantly

1.4 Business Justification

Consolidating student records, attendance, and results onto a single platform reduces administrative overhead, improves data accuracy, and enables faculty and administrators to make timely, data-driven decisions. It also improves the student experience by giving learners self-service access to their own academic information.

2. Design Thinking Report

Design Thinking is an iterative, human-centered problem-solving framework that prioritizes deep empathy for users before evaluating technological constraints. The five stages below were applied to the Student Record and Academic Management Platform.



Stage 1: Empathize

The primary user groups observed in their natural academic workflow are: **Students, Faculty Members, Academic Administrators, Examination Cell Staff**, and **IT Support Personnel**. Each group experiences distinct friction points when interacting with manual or fragmented record-keeping systems.

Stage 2: Define

Core Problem Statement

"Students and academic staff experience significant delays and errors in registering for courses, recording attendance, and accessing examination results due to fragmented, manual record-keeping. Administrators lack a centralized, real-time view of academic data, resulting in reporting delays, data inconsistencies, and reduced institutional efficiency."

Stage 3: Ideate

Unconstrained brainstorming produced the following high-value feature ideas:

- Self-service student portal for registration, attendance, and results
- Automated attendance eligibility and shortage alerts
- Real-time result and grade computation engine
- Role-based administrative dashboards with exportable reports
- Notification system for exam schedules, deadlines, and announcements

Stage 4: Prototype

Low-fidelity wireframes were conceptualized for the core UI modules: **Student Login & Dashboard, Course Registration Screen, Faculty Attendance & Marks Entry Console**, and **Admin Analytics Panel**.

Stage 5: Test

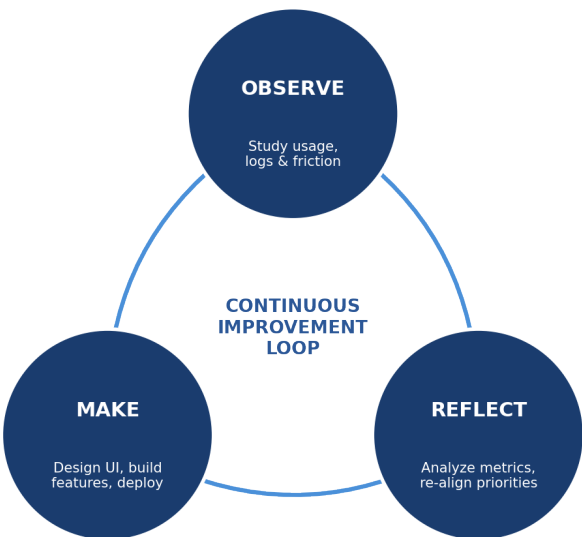
Prototype flows were reviewed against representative student, faculty, and administrator scenarios to validate navigation clarity, data entry efficiency, and report accessibility prior to backend engineering.

3. IBM Design Thinking Report & IBM Loop Documentation

3.1 IBM Design Thinking Pillars

Pillar	Application to This Platform
Focus on User Outcomes	Feature goal → “Build a student attendance table.” Outcome focus → “Enable students to view real-time attendance eligibility in under 3 clicks.”
Restless Reinvention	Treat continuous deployment as an ongoing refinement cycle — continuously analyzing usage data to improve registration flow and dashboard response time.
Diverse Empowered Teams	Cross-functional collaboration across Product Owners, UX Designers, Software Engineers, QA, Database Administrators, and DevOps Engineers.

3.2 The IBM Loop: Observe → Reflect → Make



This cycle is applied continuously across sprints: usage data from registration and attendance modules is observed, reflected upon against institutional KPIs, and translated into refined features and fixes.

4. Stakeholder Analysis Report

4.1 Identified Stakeholders

- **Students** — primary end users registering for courses and viewing academic records.
- **Faculty Members** — record attendance, enter marks, and communicate with students.
- **Academic Administrators** — oversee institutional academic operations and compliance.
- **Examination Cell Staff** — manage exam scheduling, result processing, and transcripts.
- **IT Support Personnel** — maintain platform uptime, security, and technical support.

4.2 Stakeholder Responsibility Matrix

Stakeholder Group	Key Roles	Responsibilities & Project Expectations
Business Stakeholders	College Management, Academic Administrators, Students, Faculty	Define business vision, validate functional usability, ensure academic policy compliance.
Engineering Stakeholders	Software Developers, UI/UX Designers, QA Engineers	Architect the codebase, build responsive interfaces, write automated tests, ensure quality.
Operations Stakeholders	DevOps Engineers, Database Administrators, Security Team	Provision infrastructure, maintain database integrity, automate CI/CD pipelines and monitoring.
Examination Cell	Exam Controller, Result Processing Staff	Validate examination workflows, result computation rules, and transcript formats.

5. User Persona Report

Each persona below includes explicit goals, day-to-day responsibilities, and expectations from the platform, which directly informed the functional requirements defined in Section 6.



Ananya — Student

Goals: Register for courses on time; track attendance and results without visiting the office.

Frustrations: Long queues for registration; delayed result announcements; no visibility into attendance status.

Key Feature: Self-service portal: course registration, attendance tracker, digital mark sheets.



Prof. Mehta — Faculty

Goals: Record attendance and marks quickly; communicate updates to students.

Frustrations: Manual attendance registers; duplicate data entry across systems; delayed grade submission deadlines.

Key Feature: Digital attendance & marks-entry console with bulk upload and validation.



Dr. Kapoor — Academic Administrator

Goals: Monitor institutional academic performance; generate compliance and accreditation reports.

Frustrations: No real-time dashboard; manual report compilation from multiple departments takes days.

Key Feature: Executive analytics dashboard with one-click report export.

6. Functional Requirements Specification (FRS)

- **FR-01 Authentication:** Secure role-based access control (RBAC) for Admin, Faculty, Student, and Examination Cell roles.
- **FR-02 Student Registration:** Full CRUD operations for student profiles, program enrollment, and academic year assignment.
- **FR-03 Course Registration Engine:** Semester-wise course selection with prerequisite and seat-limit validation.
- **FR-04 Attendance Management:** Session-wise attendance capture, aggregation, and eligibility threshold calculation.
- **FR-05 Examination & Result Management:** Marks entry, automated grade computation, and transcript generation.
- **FR-06 Reporting Engine:** Configurable academic and administrative reports exportable to PDF/Excel.
- **FR-07 Notification System:** Automated alerts for registration deadlines, attendance shortages, and result publication.

7. Non-Functional Requirements Specification (NFRS)

- **NFR-01 Security:** Password hashing (Bcrypt/Argon2), JWT-based session management, and HTTPS encryption for all traffic.
- **NFR-02 Performance:** Sub-500ms API response time for search and dashboard queries under concurrent user load.
- **NFR-03 Reliability & Availability:** Containerized deployment targeting 99.9% uptime with automated failover.
- **NFR-04 Scalability:** Horizontal scaling to support institution-wide concurrent usage during peak periods (registration, result day).
- **NFR-05 Usability:** Responsive, accessible interface usable across desktop and mobile devices with minimal training.
- **NFR-06 Maintainability:** Modular codebase with Infrastructure as Code (IaC) for consistent, repeatable deployments.

8. Product Vision Document

8.1 Vision Statement

"To deliver a centralized, secure, and intuitive Student Record and Academic Management Platform that eliminates manual administrative overhead, streamlines registration and attendance workflows, and gives every student, faculty member, and administrator real-time access to accurate academic information."

8.2 Mission Statement

To empower higher-education institutions with a reliable, scalable digital platform that replaces fragmented manual processes with a single, trustworthy source of academic truth.

8.3 Success Criteria

- 99.9% accuracy in student records with zero unrecorded duplicates
- Attendance and result processing time reduced by over 80%
- Sub-500ms response time for core search and dashboard operations
- Positive adoption feedback from students, faculty, and administrators within the first semester

8.4 Business Objectives

- Reduce administrative workload associated with manual record-keeping
- Improve transparency and self-service access for students and faculty
- Provide administrators with real-time, data-driven institutional insights
- Establish a scalable technical foundation for future academic modules (fees, hostel, library, etc.)

9. Minimum Viable Product (MVP) Definition

To maintain agility, non-essential capabilities (such as AI-based academic recommendations or third-party payment gateway integration) are deferred to later releases. The MVP (Sprint 1–4) focuses on core academic operations:

- Student & Faculty Login with Role-Based Access Control
- Student Registration & Profile Management
- Course Registration Module
- Attendance Capture & Basic Eligibility Tracking
- Examination Result Entry & Mark Sheet Generation
- Basic Administrative Reporting Dashboard

10. Product Roadmap

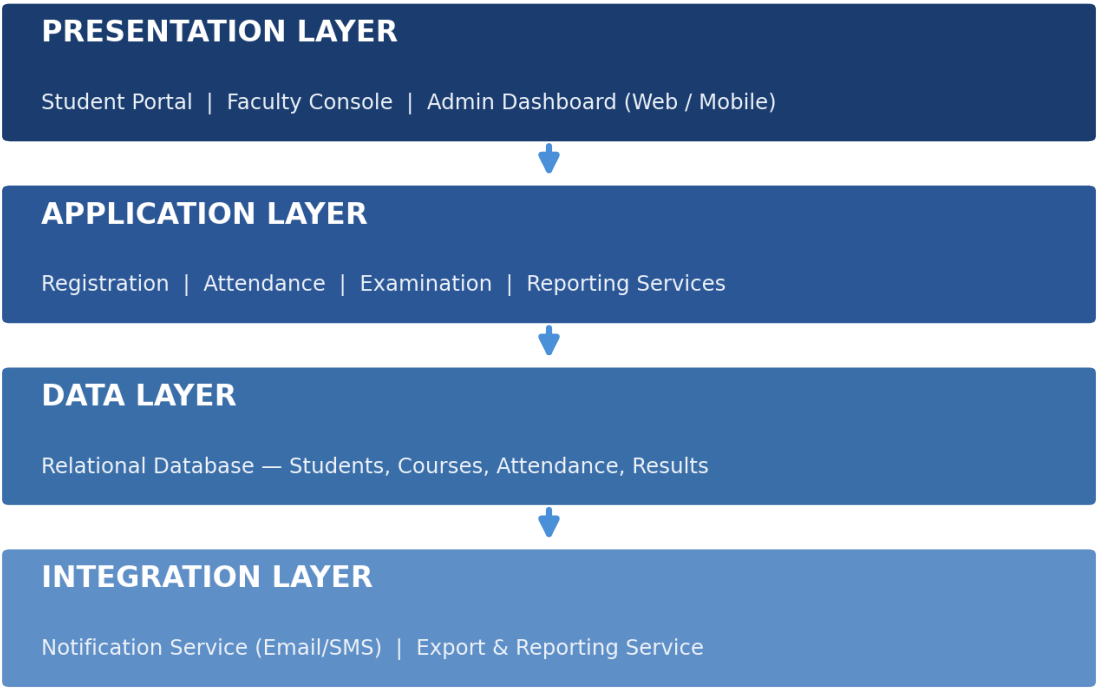


This roadmap is aligned with the progression of the training course: Unit 1 covers Phases 1–2 (this document); subsequent units address Agile planning, development, and DevOps automation.

11. Initial System Architecture Diagram

Layered Module View

The platform follows a layered architecture, separating user-facing interfaces from core business logic, data storage, and external integrations:



12. Project Charter

Field	Details
Project Name	Enterprise Student Record and Academic Management Platform
Project Purpose	Replace manual academic record-keeping with a centralized digital platform.
Sponsor	College Management / Academic Administration
Project Lead	Product Owner (Training Cohort)
Scope (Unit 1)	Business understanding, design thinking, requirements, vision, MVP, and roadmap definition.
Out of Scope (MVP)	AI-based recommendations, third-party payment gateways, hostel/library integration.
Key Milestones	Domain analysis → Design thinking → Requirements → Vision & MVP → Repository setup
Success Measures	99.9% record accuracy, 80% faster attendance/result processing, positive user adoption

13. Project Repository Initialization & Folder Structure

A Git repository is initialized for the platform with a standardized folder structure to support collaborative development in subsequent units:

```
■ student-academic-management-platform/
  ■ .github/
  ■ app/
    ■ modules/ (Students, Courses, Attendance, Results)
    ■ database/ (Schemas, Migrations, ORM Models)
    ■ tests/ (Unit & Integration Tests)
  ■ docs/ (Architecture, FRS / NFRS Specs)
  ■ docker/ (Dockerfiles, Compose Setup)
  ■ kubernetes/ (Deployments, Services, Ingress)
  ■ terraform/ (Infrastructure Provisioning)
  ■ ansible/ (Configuration Playbooks)
  ■ monitoring/ (Observability & Dashboards)
```

Note: README.md sits at the repository root alongside the folders shown above.

Git Initialization Steps

- `git init` — initialize the local repository
- Create the folder structure shown above
- `git add . && git commit -m "Initial project structure"`
- Connect to a remote repository and push the initial commit

14. Product Discovery Documentation & Deliverables Checklist

The following deliverables were produced during Unit 1 as part of the product discovery and planning process, in preparation for Agile sprint planning in subsequent units.

- **[Completed]** Business Requirements Document (BRD)
- **[Completed]** Product Vision Document
- **[Completed]** Stakeholder Analysis Report
- **[Completed]** User Persona Report
- **[Completed]** Functional Requirements Specification (FRS)
- **[Completed]** Non-Functional Requirements Specification (NFRS)
- **[Completed]** Design Thinking Report
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- **[Completed]** MVP Definition
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- **[Completed]** Project Charter
- **[Completed]** Project Repository Initialization
- **[Completed]** Project Folder Structure
- **[Completed]** Product Discovery Documentation

End of Unit 1 Mini Project Report — Enterprise Student Record and Academic Management Platform.